

## Calc 1 – The Derivative

### *Worksheet A*

1. Use the limit definition to compute  $f'(x)$  for  $f(x) = 3x^2 - x$ .
2. Find the tangent line to  $y = \sqrt{x}$  at  $x = 4$ .
3. For  $f(x) = |x - 2|$ , decide whether  $f'(2)$  exists. Justify.
4. Compute  $f'(1)$  where  $f(x) = x^3 - 2x$  using the limit definition (at the point).
5. The position of a particle is  $s(t) = t^3 - 6t^2 + 9t$ . Find its velocity and acceleration; when is the particle momentarily at rest?

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### *Worksheet B*

1. Use the limit definition to compute  $f'(x)$  for  $f(x) = 1/x$ .
2. Find the equations of both the tangent and the normal line to  $y = x^2 + 1$  at  $x = 1$ .
3. Show  $f(x) = x^{1/3}$  has a vertical tangent at  $x = 0$ .
4. For  $f(x) = \begin{cases} x^2, & x \leq 1 \\ 2x - 1, & x > 1 \end{cases}$ , is  $f$  differentiable at  $x = 1$ ?
5. A balloon rises so its height (m) at time  $t$  (s) is  $h(t) = 5t + t^2$ . Find its velocity at  $t = 3$  and the average velocity on  $[1, 3]$ .

**Calc 1 – The Derivative**  
*Answer Key (Worksheets A & B)*

**Worksheet A.**

1.  $\frac{3(x+h)^2 - (x+h) - (3x^2 - x)}{h} = \frac{6xh + 3h^2 - h}{h} = 6x + 3h - 1 \rightarrow \boxed{6x - 1}$ .
2.  $y'(x) = \frac{1}{2\sqrt{x}}$ , so slope at  $x = 4$  is  $1/4$ ; point  $(4, 2)$ . Tangent:  $\boxed{y = 2 + \frac{1}{4}(x - 4)}$ .
3. Left slope  $\lim_{h \rightarrow 0^-} \frac{|2+h-2|}{h} = \frac{-h}{h} = -1$ ; right slope  $= +1$ . Slopes differ  $\Rightarrow f'(2)$  **does not exist**.
4.  $\lim_{h \rightarrow 0} \frac{(1+h)^3 - 2(1+h) - (-1)}{h} = \lim \frac{3h + 3h^2 + h^3 - 2h}{h} = \lim(1 + 3h + h^2) = \boxed{1}$ .
5.  $v(t) = 3t^2 - 12t + 9 = 3(t-1)(t-3)$ ,  $a(t) = 6t - 12$ . Particle at rest when  $v = 0$ :  $\boxed{t = 1, 3}$ .

**Worksheet B.**

1.  $\frac{1/(x+h) - 1/x}{h} = \frac{-h}{hx(x+h)} = \frac{-1}{x(x+h)} \rightarrow \boxed{-1/x^2}$ .
2.  $y'(1) = 2$ . Tangent:  $y = 2 + 2(x - 1)$ . Normal slope  $-1/2$ :  $y = 2 - \frac{1}{2}(x - 1)$ .
3.  $f'(x) = \frac{1}{3}x^{-2/3} \rightarrow +\infty$  as  $x \rightarrow 0$ . So tangent at  $(0, 0)$  is vertical.
4. Continuity: both pieces give 1 at  $x = 1$ . Left derivative  $= 2x|_{x=1} = 2$ ; right derivative  $= 2$ . Equal  $\Rightarrow$  **differentiable**;  $f'(1) = 2$ .
5.  $h'(t) = 5 + 2t$ , so  $h'(3) = \boxed{11}$  m/s. Avg velocity  $= \frac{h(3) - h(1)}{3 - 1} = \frac{24 - 6}{2} = \boxed{9}$  m/s.